

Course: B.Tech

Branch : Common to All

Subject Code &amp; Name: 24AF1000ES106A - PROGRAMMING FOR PROBLEM SOLVING

Semester :I

Max Marks: 60

Date:24/01/2026

Duration: 3 Hr.

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                          | (Level/CO)      | Marks     |
|-------------|--------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                                   |                 | <b>12</b> |
| 1           | Which data type is used to store decimal numbers in C?<br>a) int<br>b) char<br>c) float<br>d) void                       | <b>Remember</b> | 1         |
| 2           | Which statement is used to skip the current iteration of a loop?<br>a) break<br>b) exit<br>c) continue<br>d) return      | <b>Remember</b> | 1         |
| 3           | Which symbol is used to indicate the address of a variable?<br>a) *<br>b) &<br>c) #<br>d) @                              | <b>Remember</b> | 1         |
| 4           | Which of the following is a library function in C?<br>a) add ( )<br>b) multiply ( )<br>c) printf ( )<br>d) factorial ( ) | <b>Remember</b> | 1         |
| 5           | Which of the following is an output device?<br>a) Mouse<br>b) Keyboard<br>c) Scanner<br>d) Printer                       | <b>Remember</b> | 1         |
| 6           | Which operator is used to compare two values for equality?<br>a) =<br>b) !=<br>c) ==<br>d) <=                            | <b>Remember</b> | 1         |

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| 7                                           | A string in C is terminated by which character?<br>a) newline (\n)<br>b) space ( )<br>c) null character (\0)<br>d) tab (\t)                            | Remember   | 1         |
| 8                                           | Which keyword is used to define a union?<br>a) struct<br>b) typedef<br>c) union<br>d) enum                                                             | Remember   | 1         |
| 9                                           | Which of the following is used to close a file in C?<br>a) close ( )<br>b) endfile ( )<br>c) fclose ( )<br>d) fileclose ( )                            | Remember   | 1         |
| 10                                          | Which file mode is used to append data to an existing file?<br>a) r<br>b) w<br>c) a<br>d) r+                                                           | Remember   | 1         |
| 11                                          | Which pointer stores the address of another pointer?<br>a) null pointer<br>b) dangling pointer<br>c) pointer to pointer<br>d) void pointer             | Remember   | 1         |
| 12                                          | Which unit performs arithmetic and logical operations in a computer?<br>a) Control Unit<br>b) Memory Unit<br>c) Arithmetic Logic Unit<br>d) Input Unit | Remember   | 1         |
| <b>Q. 2 Solve the following.</b>            |                                                                                                                                                        |            | <b>12</b> |
| A)                                          | Explain the different phases of the C program execution process. Also describe syntax errors with examples.                                            | Understand | 6         |
| B)                                          | Describe the components of computer memory hierarchy. Explain primary and secondary memory with a neat diagram.                                        | Understand | 6         |
| <b>Q.3 Solve the following.</b>             |                                                                                                                                                        |            | <b>12</b> |
| A)                                          | Write a C program to find whether a given number is positive, negative, or zero.                                                                       | Apply      | 6         |
| B)                                          | Explain increment and decrement operators. Also explain assignment operators with suitable examples.                                                   | Understand | 6         |
| <b>Q. 4 Solve Any Two of the following.</b> |                                                                                                                                                        |            | <b>12</b> |
| A)                                          | Explain the concept of recursion in C with advantages and disadvantages.                                                                               | Understand | 6         |

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| <b>B)</b> | Write a C program using a user-defined function to check whether a number is a palindrome. (Note: A palindrome is a value (number, word, or string) that reads the same forward and backward. e.g. 1331, 4554 ) | <b>Apply</b> | <b>6</b> |
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| <b>C)</b> | Explain storage classes in C. | <b>Understand</b> | <b>6</b> |
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| <b>Q.5</b> | <b>Solve Any Two of the following.</b> |  | <b>12</b> |
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| <b>A)</b> | Write a C program to sort an array of integers using bubble sort. | <b>Apply</b> | <b>6</b> |
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| <b>B)</b> | Explain string handling functions in C with examples. Write a program to concatenate two strings. | <b>Apply</b> | <b>6</b> |
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| <b>C)</b> | Explain dynamic memory allocation using malloc( ), calloc( ), realloc( ) with examples. | <b>Apply</b> | <b>6</b> |
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| <b>Q. 6</b> | <b>Solve Any Two of the following.</b> |  | <b>12</b> |
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| <b>A)</b> | Write a C program to copy the contents of one file into another file. | <b>Apply</b> | <b>6</b> |
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| <b>B)</b> | Explain structures and unions. Compare structures and unions with examples. | <b>Understand</b> | <b>6</b> |
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| <b>C)</b> | Explain looping statements in C:<br>a) while loop<br>b) do-while loop | <b>Understand</b> | <b>6</b> |
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**\*\*\* End \*\*\***